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Background

- Biologic agents and small molecule therapies have transformed the treatment of inflammatory autoimmune conditions like rheumatoid arthritis, psoriatic arthritis, and psoriasis.
- While employing a variety of mechanisms to inhibit unwanted immune responses, these agents are also associated with an increased risk of infection, particularly respiratory tract infections.¹
- Clinicians are left to manage these potential side effects and often recommend withholding immunosuppressive therapy during active infections, especially when patients are febrile or receiving antibiotic treatment, to minimize complications and expedite recovery.²
- Health-system specialty pharmacy (HSSP) pharmacists are in a unique position to identify infections and/or infection risk.
- Patients serviced by HSSP pharmacists can easily contact HSSPs to ask questions regarding infection management while being treated for these inflammatory conditions.
- HSSP pharmacists likely have a positive impact on helping patients manage these therapies during times of suspected or proven infection.

Objectives

- **Primary Objective:**
 - Evaluate the frequency of pharmacist interventions related to infection management in patients receiving biologic or small molecule therapies for inflammatory autoimmune conditions at a health-system specialty pharmacy

Methods

- **Design:** Retrospective chart review of interventions performed by HSSP pharmacists
- **Setting:** West Virginia University Medicine Specialty Pharmacy
- **Timeline:** October 1, 2024 - December 31, 2024
- **Inclusion Criteria:**
 - Diagnosis of psoriasis, psoriatic arthritis, rheumatoid arthritis, spondyloarthritis, and conditions specified as “other inflammatory conditions”
 - Intervention mentioned any suspected or confirmed infection
 - Enrolled in the patient management program

Results

Demographics (n=136)

Female	79.4% (108)
Age	44 (17-78)
Diagnosis	
Spondyloarthritis	7.4% (10)
Psoriasis	9.6% (13)
Psoriatic arthritis	16.9% (23)
Other inflammatory conditions	26.5% (36)
Rheumatoid arthritis	39.7% (54)

Table 1: Patients receiving specialty medications between October 1st, 2024, and December 31st, 2024

Pharmacist intervention types and recommendations (n=178)

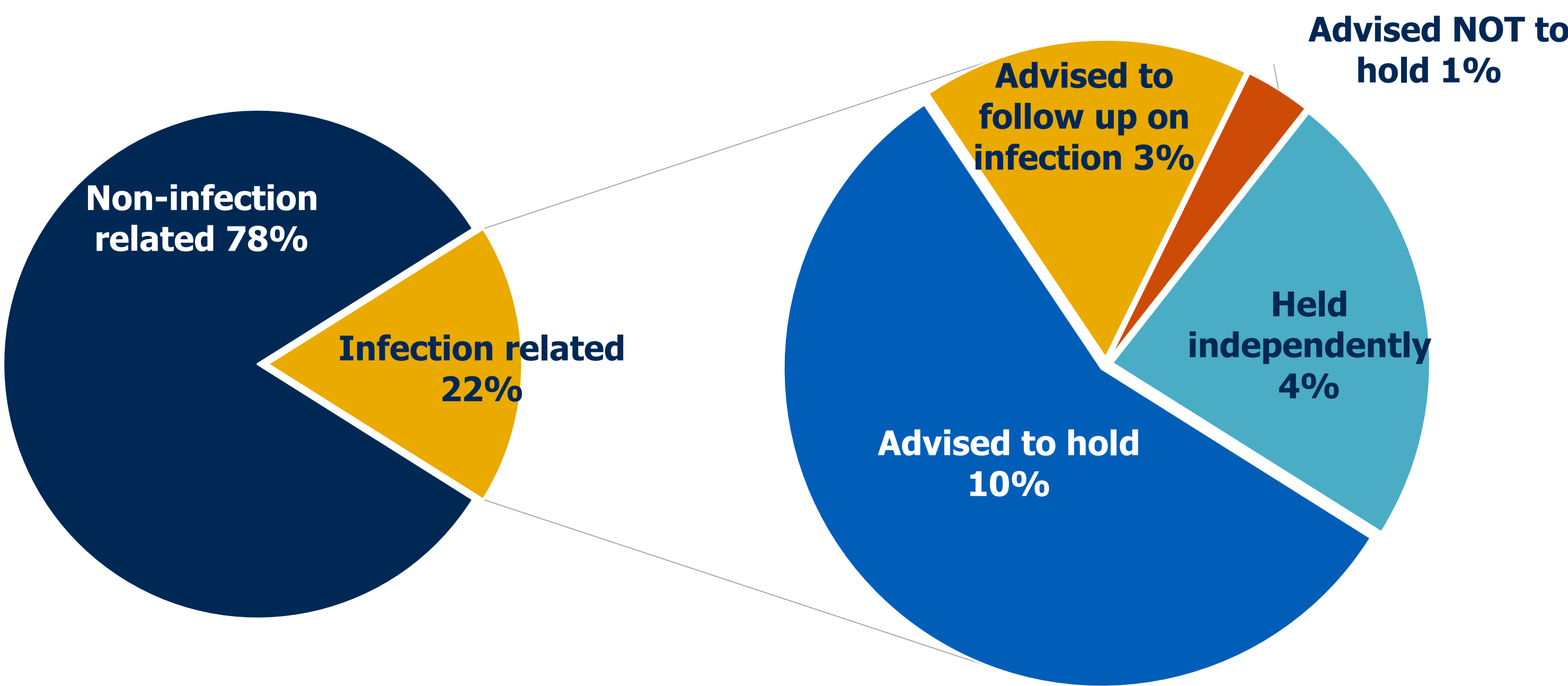


Chart 1: Pharmacist interventions for HSSP patients receiving specialty medications between October 1st, 2024, and December 31st, 2024

100% of pharmacist recommendations were accepted

Disease State	Interventions
Other inflammatory conditions	20% (6)
Psoriasis	7% (2)
Psoriatic arthritis	17% (5)
Rheumatoid arthritis	43% (13)
Spondyloarthritis	13%(4)
Grand Total	30

Table 2: Incidence of intervention by disease state

Results

Drug class	Interventions by patient
Tumor Necrosis Factor Inhibitor	36.7% (11)
Janus Kinase Inhibitor	20% (6)
Interleukin-17 Inhibitor	23.3% (7)
Interleukin-6 Inhibitor	13.3% (4)
T-cell Modulator	3.3 % (1)
Antimetabolite	3.3 % (1)
Grand Total	30 patients

Table 3: Incidence of intervention by drug class

Infection related interventions across disease type and drug class (n=30)

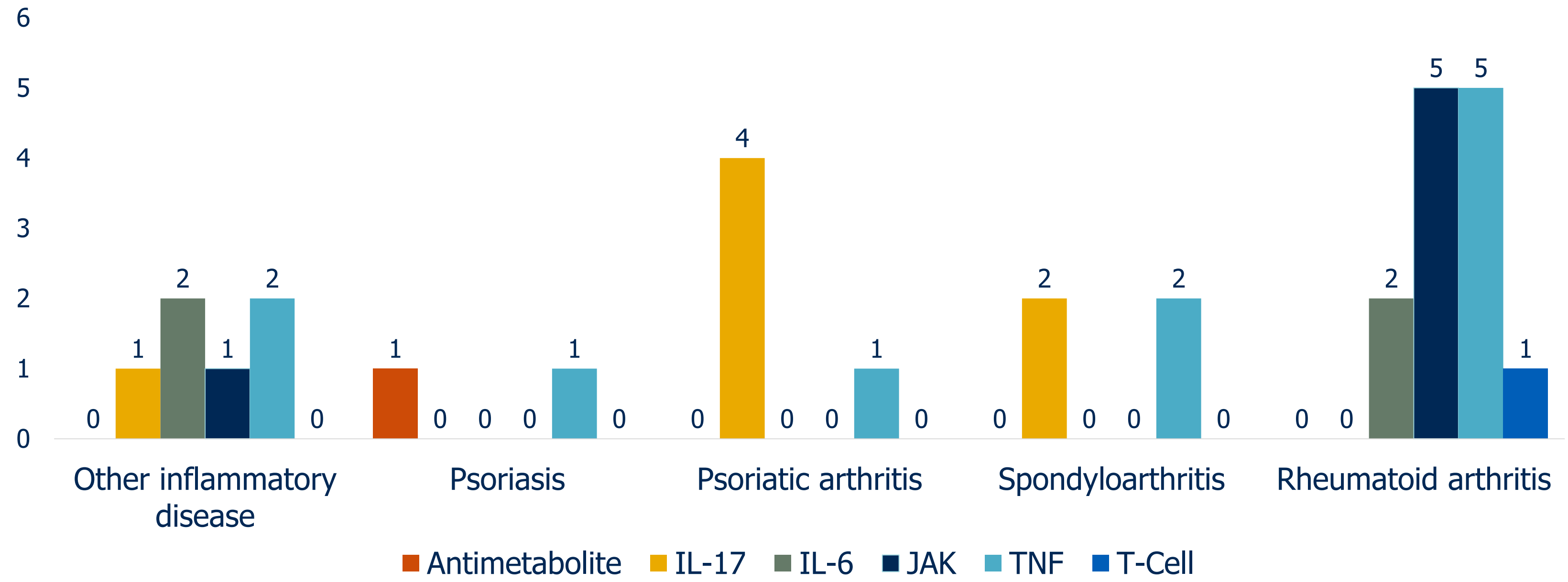


Chart 2: Incidence of intervention by drug class and disease state

Conclusions

- This study highlights the key role of specialty pharmacists in managing patients being treated with biologic or small molecule therapies.
- Many interventions (22%) were related to active or suspected infection management and all the pharmacist recommendations were accepted, suggesting that pharmacists are integral to optimizing patient care and preventing infection-related complications in this population.
- The study data is limited by its short duration and lack of long-term clinical impact data points following a hold.
- Future directions include evaluating:
 - Long-term clinical impact of medication holds through collecting frequency of disease flares following holds due to infection
 - Frequency of infection necessitating hold
 - Clinical outcomes of those patients that do not hold their medications during infection

References

1. Davis JS, Ferreira D, Paige E, Gedye C, Boyle M. Infectious Complications of Biological and Small Molecule Targeted Immunomodulatory Therapies. Clinical Microbiology Reviews. 2020;33(3). doi:https://doi.org/10.1128/cmr.00035-19
2. Menter A, Strober BE, Kaplan DH, et al. Joint AAD-NPF guidelines of care for the management and treatment of psoriasis with biologics. Journal of the American Academy of Dermatology. 2019;80(4):1029-1072. doi:https://doi.org/10.1016/j.jaad.2018.11.057